Question	Answer	Marks
1	**Defining the problem**	
	vary m and measure f **or** m is the independent variable and f is the dependent variable	**1**
	keep f_0 constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • turntable base and motor on bench • two labels from turntable, motor, belt, C, putty, bench, release mechanism	**1**
	workable circuit showing terminals of motor connected to (d.c.) supply	**1**
	method to measure time to determine f and f_0, e.g. use a stopwatch to measure (n)t **and** method to measure m, e.g. use a (top-pan) balance	**1**
	measure the time t for n revolutions **and** $T = t / n$ **and** $f = 1 / T$ **or** $f = n / t$	**1**
	Method of Analysis	
	plots a graph of $\frac{1}{f}$ against m or equivalent Do not accept logarithms.	**1**

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Question	Answer	Marks
1	$\frac{1}{f}$ against m | m against $\frac{1}{f}$ $K = \frac{r^2}{f_0 \times \text{gradient}}$ | $K = \frac{r^2 \times \text{gradient}}{f_0}$	**1**
	$\frac{1}{f}$ against m | m against $\frac{1}{f}$ $\beta = f_0 \times y\text{-intercept}$ | $\beta = -\frac{f_0 \times y\text{-intercept}}{\text{gradient}}$ **or** $\beta = -\frac{r^2 \times y\text{-intercept}}{K}$	**1**

Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 precaution linked to <u>putty</u> leaving turntable, e.g. screens around apparatus / cushions to prevent putty from leaving bench	
	D2 keep *r* <u>constant</u>	
	D3 use rule(r) to measure *r*	
	D4 stand and fixed point located above P (to ensure *r* is constant)	
	D5 method to ensure putty is dropped at (constant) distance *r*, e.g.; fixed location above the turntable to drop putty to keep *r* constant **or** draw a circle of radius *r* on the turntable	
	D6 method to keep f_0 constant, e.g. adjust the rheostat to keep the current in the motor constant / check ammeter	
	D7 clamp motor and/or turntable (base) <u>to bench</u>	
	D8 lubricate the turntable to reduce friction	
	D9 repeat experiment for the same value of *m* **and** determine the average *f*	
	D10 relationship valid <u>if</u> a straight line is produced (with *y*-intercept = $\frac{\beta}{f_0}$) Do not accept line through the origin.	

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